Honeypot Project – Second Deployment & Improvements

Introduction

Following the initial deployment, I redeployed the honeypot on AWS with several improvements based on earlier observations and supervisor feedback.

The aim of this second attempt was to enhance data collection, improve realism in exposed services, and gain deeper insight into attacker behaviour. This version builds on the first deployment by refining configurations and expanding the scope of monitoring.

Improvements & Observations

In this second deployment, several improvements were made to enhance both data collection and analysis:

- Improved dashboard to capture and visualise the tools used by attackers
- Added attack chain tracking to analyse how a single attacker interacts across multiple services
- Expanded and improved exposed services, including HTTP, HTTPS, Telnet, FTP, ONVIF, and RTSP
- Worked on SSH configuration and attempted to analyse VPN/Tor endpoints (this was implemented but did not fully work as expected)

Observations

- A higher volume of attacks was observed compared to the first deployment
- New types of attacks were identified, showing more variation in attacker behaviour
- The honeypot was discovered on platforms such as Shodan and Censys, increasing visibility and incoming traffic
- Overall interaction with services was more advanced compared to the first attempt

Supervisor Feedback

After a follow-up meeting with my supervisor:

- SSH configuration was noted as improved
- A clear target was set to complete the project and focus on final data analysis

Next Steps

- Further improve SSH configuration
- Add and monitor IoT-specific ports such as MQTT
- Remove unnecessary ports to reduce noise
- Make the environment more realistic to encourage deeper attacker interaction

Screenshots:

AWS setup:

Inbound rules (13)

Manage tags | Edit inbound rules

Search

1

Name	Security group rule ID	IP version	Type	Protocol	Port range
–	sgr-075fea190af71680d	IPv4	Custom TCP	TCP	0
–	sgr-0d2b75bfdc59133f0	IPv4	Custom TCP	TCP	1883
–	sgr-052a171d6dfc77e10	IPv4	Custom TCP	TCP	21
–	sgr-082a28c440bb001b3	IPv4	Custom TCP	TCP	8080
–	sgr-03c66a3e89b815b99	IPv4	Custom TCP	TCP	5001
–	sgr-07a8024a0ddb18f50	IPv4	HTTPS	TCP	443
–	sgr-05feb5e1503caa3b1	IPv4	Custom TCP	TCP	23
–	sgr-051138eddd7cbb7fa	IPv4	HTTP	TCP	80
–	sgr-0cf2b10dce8ab090a	IPv4	Custom TCP	TCP	8000
–	sgr-0e45ef02142353b13	IPv4	Custom TCP	TCP	5900
–	sgr-0564be993ff3aef6c	IPv4	SSH	TCP	22
–	sgr-0b2e3bde71696b7e9	IPv4	Custom TCP	TCP	554
–	sgr-0619126cfdbae10e8	IPv4	Custom TCP	TCP	2222

Resource map | CIDRs | Flow logs | Tags | Integrations

Resource map Info

Show all details

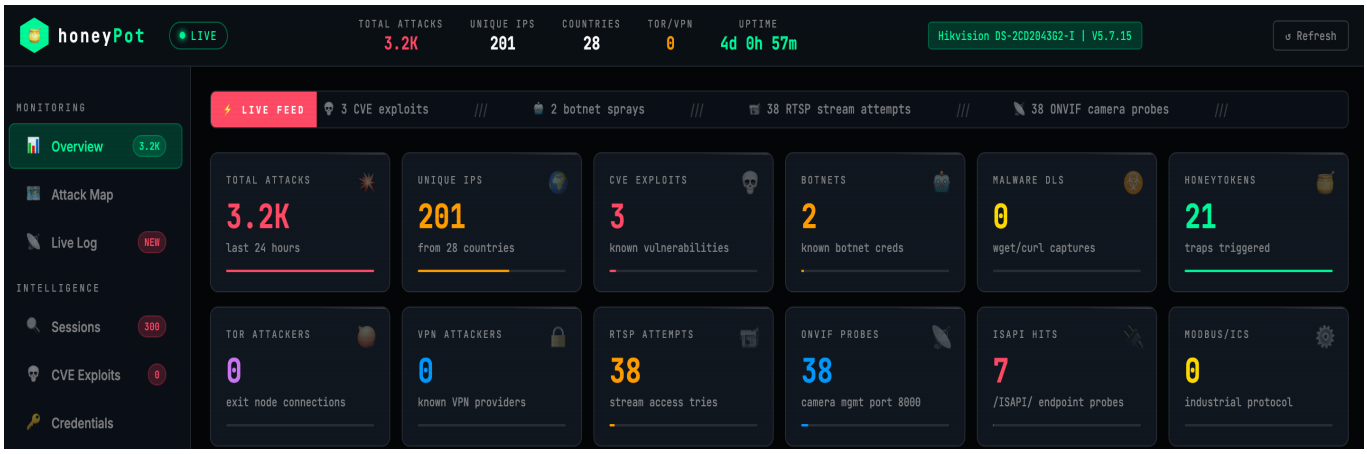
VPC
Your AWS virtual network
vpc-0594e559c78f46ccf

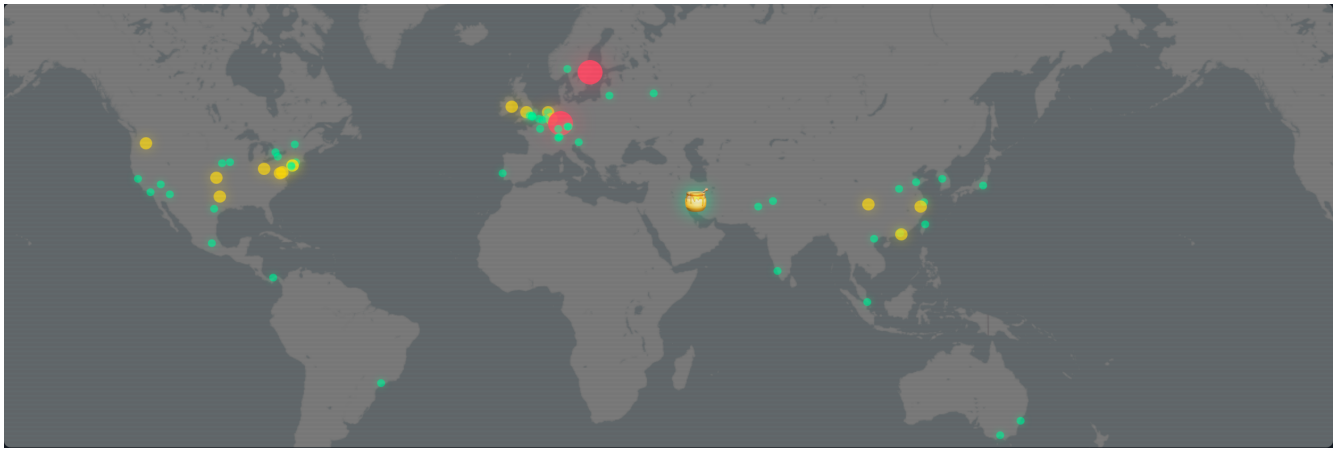
Subnets (1)
Subnets within this VPC
us-east-1a
subnet-1-1a

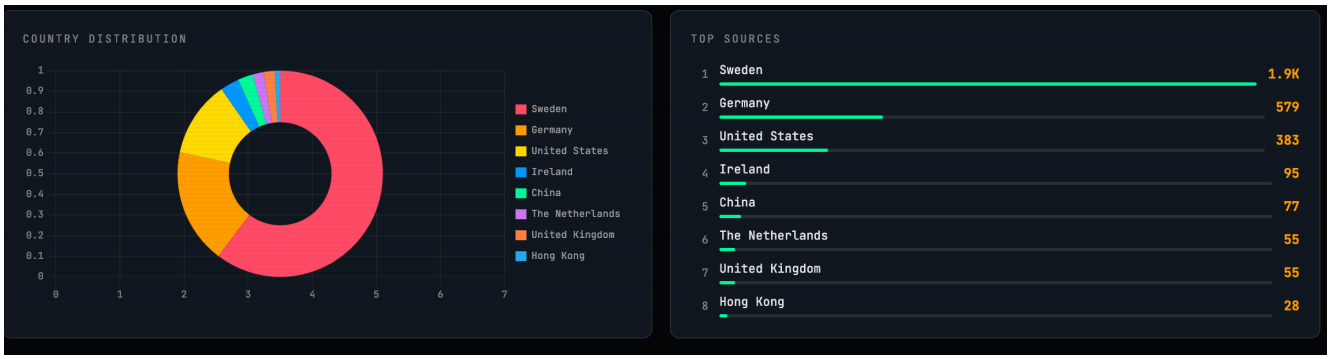
Route tables (1)
Route network traffic to resources
rtb-0a26ace707afafa60

Dashboard Result:

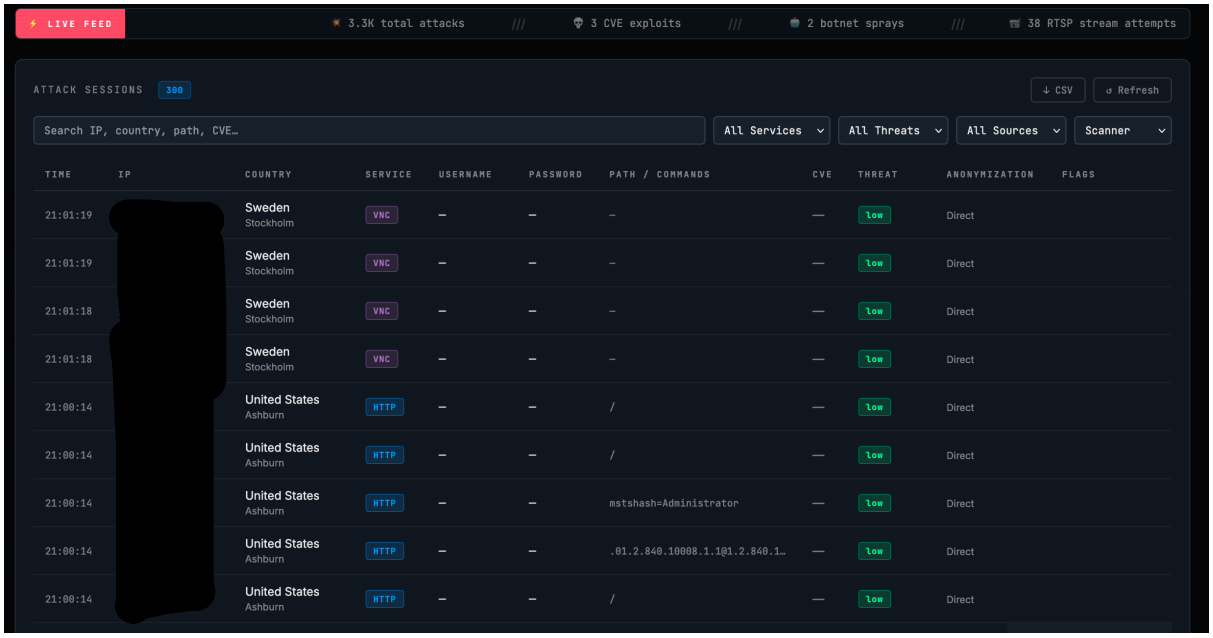
- Over a 4-day and 1-hour period, the honeypot recorded **3.2k attacks from 201 unique IP addresses**. Most of the malicious traffic originated from **Sweden**, with additional activity coming from several other regions.



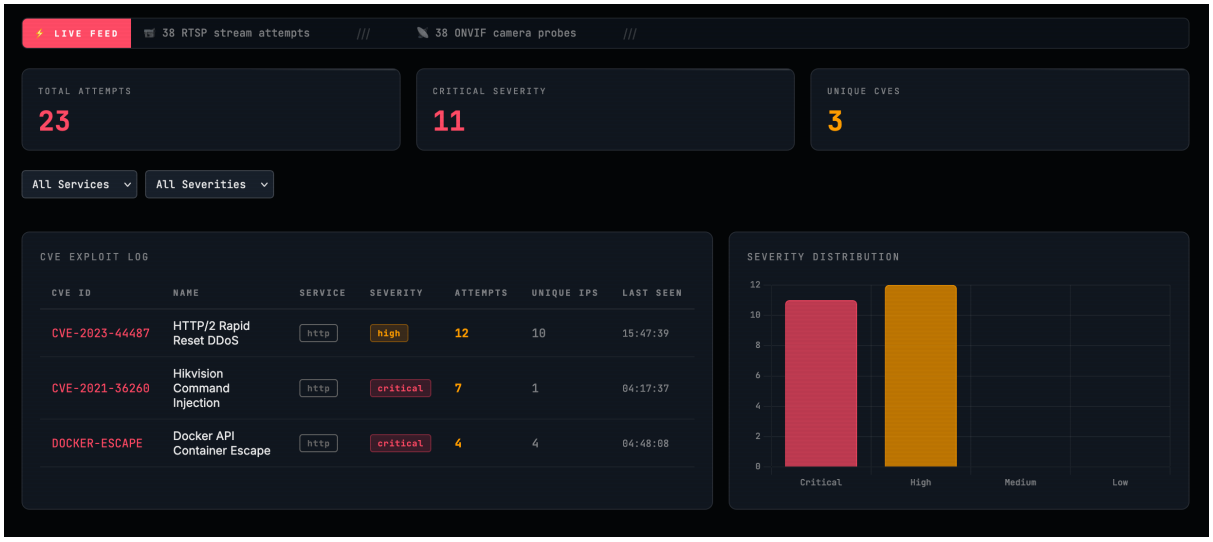




- This is the **session page**, where I can view real-time attacker interactions, including the **source IP**, services that were attacked, attempted **usernames/passwords**, and the **commands** executed on exposed services.
  The session view also highlights whether any **CVE exploits** were attempted during the connection.
  Additionally, the **flag indicator** shows if an attacker used known **botnet-related credentials**, helping identify automated malware activity.



- This section shows **how many times CVEs were exploited** during the deployment and **which specific services** the vulnerabilities were attempted against.



- During the attacks, a wide range of **usernames and passwords** were attempted by attackers

TOP CREDENTIAL PAIRS

#	USERNAME	PASSWORD	COUNT	BOTNET
1	admin	(empty)	12	BOTNET
2	GET / HTTP/1.1 Host: 52.54.209.63:23 User-Agent: visionheight.com/scan Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) Chrome/126.0.0.0 Safari/537.36 Accept: */* Accept-Encoding: gzip	(empty)	8	normal
3	admin	admin	7	BOTNET
4	anonymous	(empty)	6	normal
5	root	(empty)	6	normal
6	yoo	(empty)	5	normal
7	anonymous	anonymous@	4	normal
8	MGLNDD_52.54.209.63_23	(empty)	3	normal
9	skjdvds	acasc	2	normal
10	yoo	sdjc	2	normal
11	{wo+Q@jQ"G\G7G2HH/+ /5 4	(empty)	1	normal
12	{woA.7ئBY Sn*ö@/+ /5 4	(empty)	1	normal
13	{w j{x,d'F_://+ /5 4	(empty)	1	normal
14	{w?j\-)'@x|Lnش/*t/+ /5 4	(empty)	1	normal
15	{w?zK8	(empty)	1	normal
16	{wNEen zBNf}|&D'Iw/+ /5 4	(empty)	1	normal
17	{wd= daD&.voq/+ /5 4	(empty)	1	normal
18	{wu3w,h.쎈'o;/+ /5 4	(empty)	1	normal
19	0r@MJp6F?m:o <*KLRBxo3g9kE #+ $,rs¦/ (0`avw./<5=A 52.54.209.63	#<:hqh2ch2spdy/3spdy/2spdy/1http/1.1http/1.0http/0.9 3&$ "ivS;y1V?9EAe>d=@9-+	1	normal
20	% !"'	P@55word!	1	normal
21	% !"'	password	1	normal
22	123456	(empty)	1	normal

TOTAL ATTACKS	UNIQUE IPS	COUNTRIES	TOR/VPN	UPTIME
3.3K	202	28	0	4d 1h 1m

23	%2Fetc%2Fpassw HTTP/1.1 Host: 52.54.209.63:23 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:146.0) Gecko/20100101 Firefox/146.0 Accept: */* Connection: close	(empty)	1	normal
24	SSH-2.0-Go	L?*qqcurve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,dif	1	normal
25	SSH-2.0-Go L50K+curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-	nistp521,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1[ssh-rsa-cert-v01@openssh.com,ssh-dss-cert-v01@openssh.com,ecdsa-sha2-nistp256-cert-v01@openssh.com,ecdsa-sha2-nistp384-cert-v01@openssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.com,ssh-ed25	1	normal
26	SSH-2.0-Go L'Yy*xK Ycurve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-	nistp521,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1[ssh-rsa-cert-v01@openssh.com,ssh-dss-cert-v01@openssh.com,ecdsa-sha2-nistp256-cert-v01@openssh.com,ecdsa-sha2-nistp384-cert-v01@openssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.com,ssh-ed25	1	normal

- There were several **honeytokens triggered** during the attacks, indicating that attackers accessed or attempted to interact with decoy credentials and files designed to detect unauthorized activity.
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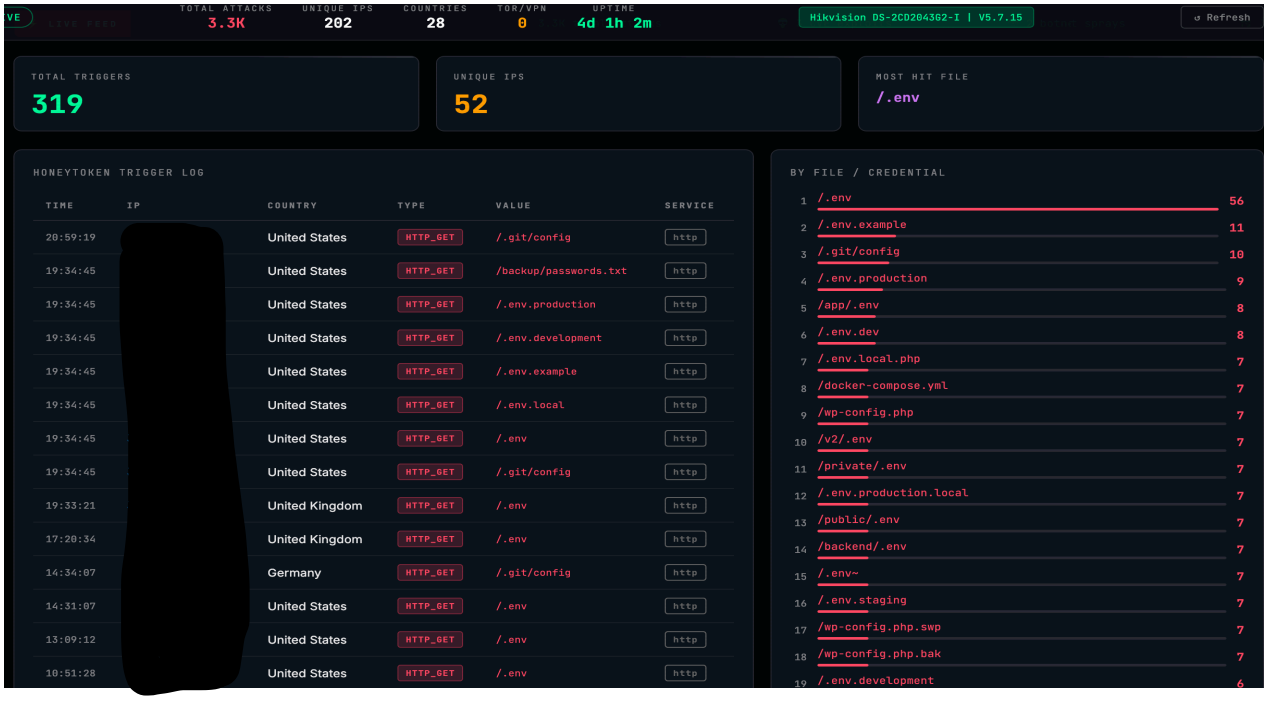
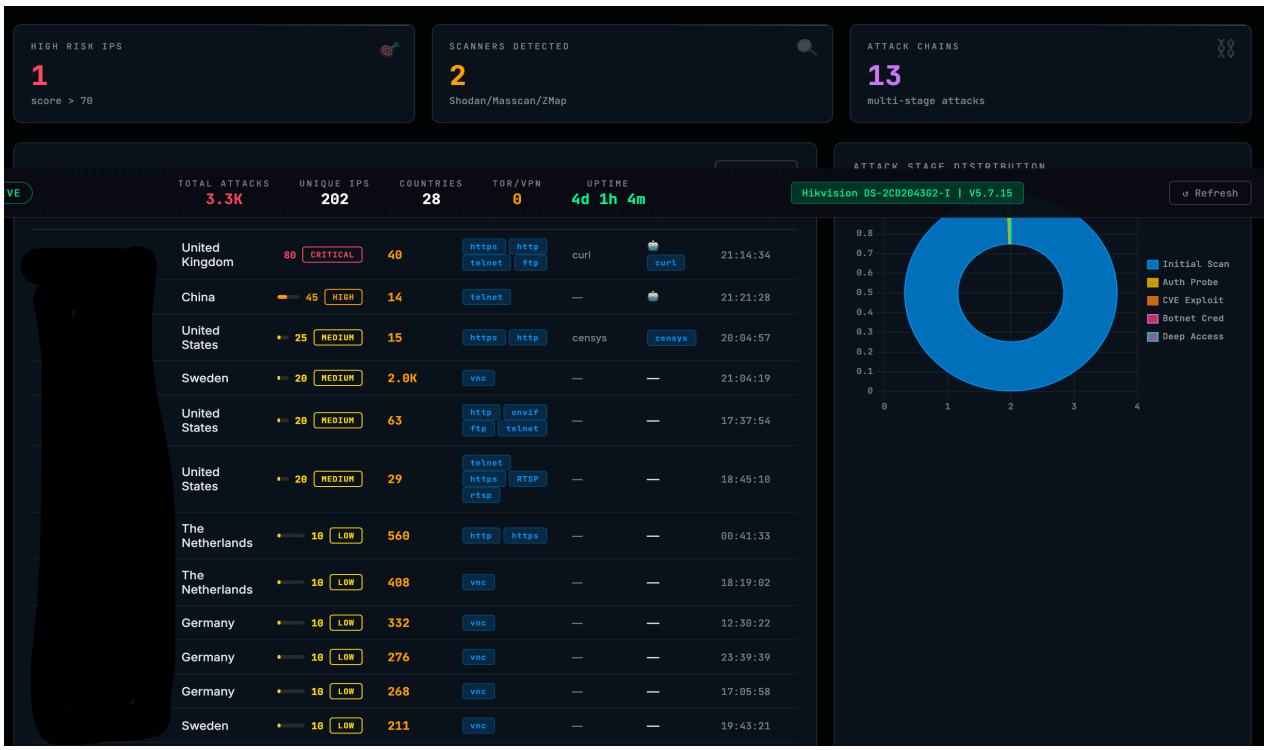
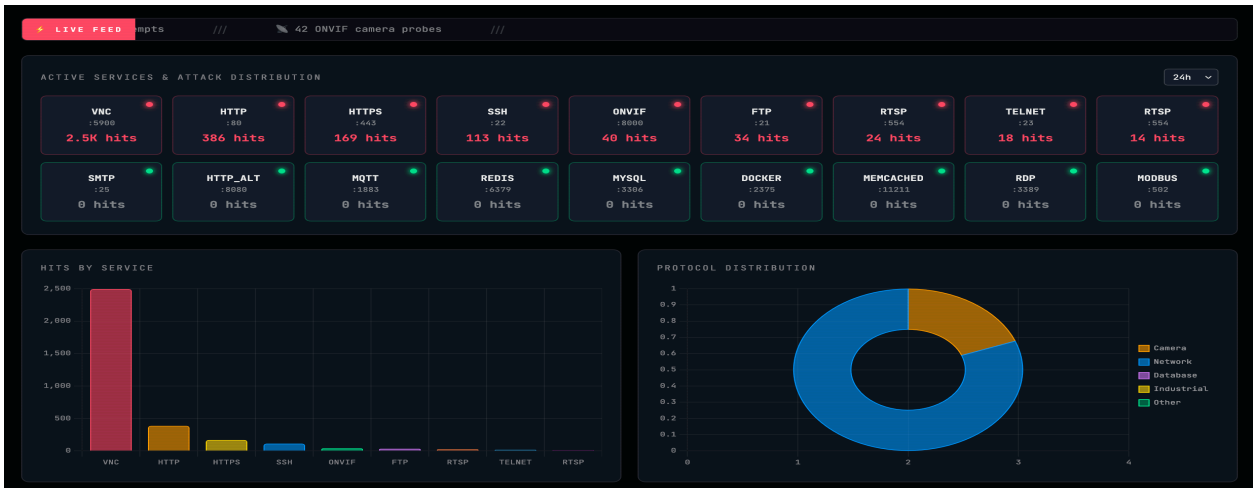


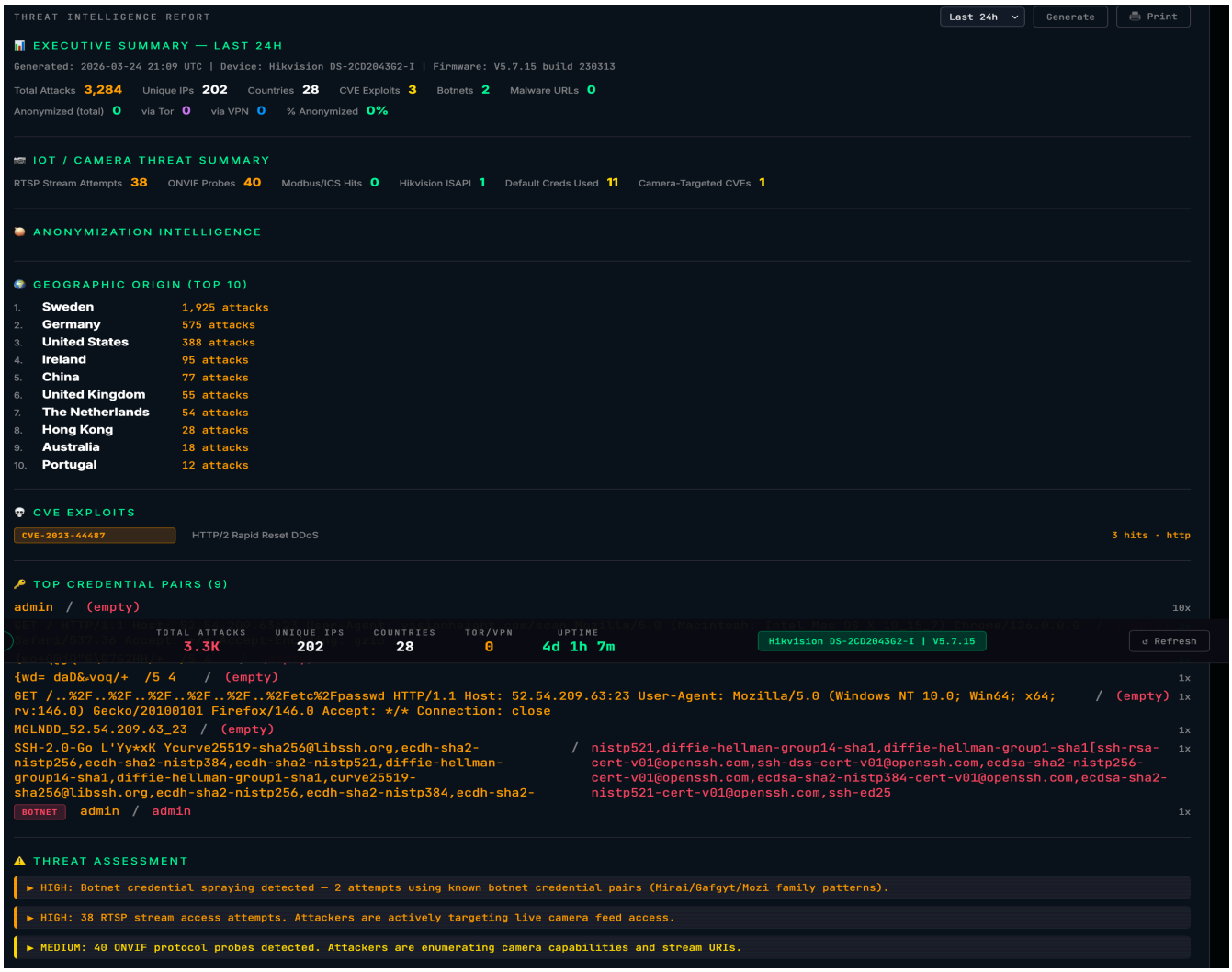
- This allows analysis of how attackers discover the honeypot, the tools and commands they use, and their behaviour during attacks.
  Attack chain tracking shows how a single IP targets multiple services across the environment.



- Again, VPN-related services were the most targeted.
  This shows which services were attacked and how many times each port was targeted.



- This is the overall summary of our honeypot project



Search Result from Shodan and Cynsys:

// TAGS: cloud

// LAST SEEN: 2026-03-23

General Information

Hostnames	ec2-52-54-209-63.compute-1.**amazonaws.com**
Domains	amazonaws.com
Cloud Provider	**Amazon**
Cloud Region	**us-east-1**
Cloud Service	**EC2**
Country	**United States**
City	**Ashburn**
Organization	**Amazon Technologies Inc.**
ISP	**Amazon.com, Inc.**
ASN	**AS14618**

Open **Ports**

80 443

// **80** / TCP 1674704659 | 2026-03-23T07:19:27.789890

IP Camera Web Manager

```
HTTP/1.1 200 OK
Server: App-webs/
Content-Type: text/html
Content-Length: 543
Connection: close
```

// **443** / TCP 586257981 | 2026-03-01T20:03:23.351580

```
HTTP/1.1 403 Forbidden
Date: Sun, 01 Mar 2026 20:03:23 GMT
Content-Type: application/json
Content-Length: 136
Connection: keep-alive
x-amzn-requestid: 8c91d123-af35-45a7-8d91-9f6f3e4e4101
access-control-allow-origin: *
```

RESULTS: **1** • DURATION: **0.07s**

Personalize Results

[redacted] • HOST

ec2-52-54-209-63.compute-1.amazonaws.com

CAMERA IOT

Network (AS)	AMAZON-AES - Amazon.com, Inc. (14618)
Location	Ashburn, Virginia (US)
Services (4)	80 / HTTP 554 / RTSP 443 / HTTP 8080 / HTTP

MATCHED FIELDS

host.ip [redacted]